

Mach-Nix and Shell Environments

Rohit Goswami · 2022-01-17 · nix · python · cmdline

I often need to set up quick virtual environments. Unfortunately, the standard approach to work with this in `nix` deals with the local `nixpkgs` mechanism for `python` dependencies:

```
nix-shell -p "python38.withPackages(ps: [ ps.numpy ps.sh ])"
```

However there is a catch for packages which are not officially present upstream.

```
# Fails!
nix-shell -p "python38.withPackages(ps: [ ps.numpy ps.sh ps.lieer ])"
```

However, the `mach-nix` project can indeed be used to work around this, at the cost of a somewhat longer command.

```
# Works
nix-shell -p '(callPackage (fetchTarball https://github.com/DavHau/mach-nix/tarball/3.0.2)
{python="python38";}).mkPython{requirements="numpy\n lieer\n ipython";})' --command ipython
```

`mkPythonShell` does not generate an environment which can be used here, since that is not a derivation which can be built. This is most useful in the context of systems which use `asdf` or other `PATH` shim approaches.